Analysis 1: Aug-Nov 2026

16 sep 2026

Yesterday most of you did not turn up in the class. Apart from some exercise, here is one important new thing I did:

Let $\{a_n : n \geq 0\}$, $\{b_n : n \geq 0\}$ be bounded sequences and let $f(x), g(x)$ be defined by for $x \in (-1, 1)$

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

$$g(x) = \sum_{n=0}^{\infty} b_n x^n$$

Then we have : f, g are continuous functions on $(-1, 1)$ and $h(x) = f(x)g(x)$ also admits a power series expansion $x \in (-1, 1)$

$$h(x) = \sum_{n=0}^{\infty} c_n x^n.$$

Find expression for $\{c_n : n \geq 0\}$ in terms of $\{a_n : n \geq 0\}$, $\{b_n : n \geq 0\}$

And yes, I was asked by someone: is power series included in the mid-sem syllabus. Answer is YES.